Pneumatics revision summary

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Which two components are used together to create a time delay?

- A uni-directional restrictor and a reservoir

2 Which actuator is operated by a low-pressure signal from an air bleed?

- A diaphragm

3 A 32 mm cylinder works at 0.5 N/mm². Calculate the outstroke force.

- $A = \pi \times 16^2 = 804$ mm²
- $F = 0.5 \times 804 = 402$ N

4 Use the word bank to describe what happens when valve 1 (a push-button 3/2 valve) is pressed in a circuit where it pilots a 5/2 valve controlling a double-acting cylinder.

- Valve 1 is actuated and sends pilot air to the 5/2 valve.
- The 5/2 valve changes state and the piston outstrokes.

5 Complete the table: draw each symbol and say what the component does.

- 3/2 valve: 3 ports, 2 states; usually controls a single-acting cylinder
- 5/2 valve: 5 ports, 2 states; controls a double-acting cylinder
- Single-acting cylinder: air outstrokes the piston, a spring instrokes it
- Double-acting cylinder: air drives the piston both out and in
- Shuttle valve: OR control; either input sends air to the output
- Restrictor: slows the air flow
- Uni-directional restrictor: slows the air flow in one direction only
- Reservoir: with a restrictor, creates a time delay